COMMERCIAL ITEM DESCRIPTION

# REFRIGERATOR, MECHANICAL, BLOOD BANK

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers a blood bank refrigerator.

Salient characteristics:

The refrigerator shall be a self-contained blood bank refrigerator with stainless-steel, removable adjustable drawers with perforated or solid bottoms and full extension, sleeve or ballbearing slides. It shall have a stainless steel interior and exterior, and stainless steel door. The door shall be non-self-closing and utilize a positive mechanical action door fastener. The door shall require a gentle push to secure and seal the refrigerator, and shall be hinged on the right hand side.

The refrigerator shall hold a minimum of 480 plastic blood bags.

The refrigerator shall meet all the requirements for refrigeration of the American Association of Blood Banks Technical Manual, latest edition.

The refrigerator shall be equipped with at least a 1/3 horsepower, hermetically sealed, compressor.

The blood bank refrigerator shall have an automatic defrosting system. Air shall circulate at the proper cfm rate throughout the cabinet without air pockets. Operation of unit shall be vibration free.

Electronic temperature control set to maintain cabinet temperature between 35.6°F to 39.2°F, shall be provided. A temperature monitor and a seven-day recording thermometer with a 700 day supply of replaceable recorder charts shall also be provided. The temperature monitor shall have an audio-visual

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alarm system designed to alert the operator of any abnormal temperature.

The refrigerator shall operate on 115 volts, 60Hz, single phase. It shall be supplied with an UL-approved three-wire ground line cord and green dot hospital grade male plug. Where a hospital grade plug is not available, a plug suitable for use in hospital environment shall be provided.

The interior of the refrigerator shall have adequate lighting to facilitate inspection of the blood and thermometers.

Insulation shall be foamed polyurethane or fiberglass and shall be of necessary thickness and efficiency for providing the following pulldown performance.

| Ambient Temperature | Pull Down Time Hrs. (Max.) | Cycle Time (Percent) |
|---|---|---|
| 120°F | 3 | 100 |
| 100°F | 1-1/4 | 55 |
| 77°F | 1/2 | 34 |

Internal compartment temperature shall be 35.6°F.

Dielectric strength. The unit shall be capable of withstanding without breakdown, for one minute, a potential of 1240 volts, 60Hz, applied between live and dead metal parts when tested as specified herein.

Identification marking. Each refrigerator shall have legible labels marked with the following:

National Stock Number
Name or registered trademark of manufacturer
Contract or purchase order number
Wiring diagram
Electrical characteristics
Instructions for use (located so as to be readily visible during use)

Wiring diagram. The diagram shall portray each electrical component by sketch, outline, or symbol; and shall include the name of each component. The wiring diagram shall show the wiring to all components and any coding used for identification of wires.



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Underwriters Laboratories, Inc. The refrigerator shall conform to UL 471 Standard for Safety - Commercial Refrigerators and Freezers. Acceptable evidence of this shall be a UL listing mark or a certified test report from an independent laboratory acceptable to the Government.

Workmanship. The blood bank refrigerator shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One refrigerator, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 2.5 (percent defective) and an inspection level of II. The unit of product for examination purposes shall be one refrigerator as



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specified in this description.

Sampling for tests. Sampling for tests including dimensional test, shall be conducted in accordance with MIL-STD-105, with an AQL of 2.5 (percent defective) and an inspection level of II. The unit of product for test purposes shall be one refrigerator

Tests.

1. Operation. The unit shall be operated for one hour according to manufacturer's directions to verify that the controls are adjusted and functional. The unit shall be inspected for damage or leakage from any part, piping or fitting.

2. Dielectric strength. A slowly rising potential, from 0 to 1240 volts, 60 Hz, shall be applied between live and dead metal parts. The potential shall be maintained at maximum for one minute.

3. Performance. The unit shall obtain an internal compartment temperature of 35.6°F with the ambient temperatures maintained as specified herein, within the times specified, and without exceeding the percentage running times specified for at least 24 hours of steady state operation under the conditions for ambient temperature, pulldown time and cycle time specified herein.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.



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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTES: 1. The following National Stock Number is covered by this document:

4110-01-288-0280

2. Application for copies of the American Association of Blood Banks Technical Manual should be addressed to:

American Association of Blood Banks
1828 L Street, N.W.
Washington, DC 20036



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MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project No. 4110-0427

Location:
CID/REFRIGERATOR/MECHL/
BLOOD/BANK/H1B-37



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